



**Dandes
Academy**

Mastering

AI & Machine Learning Program

- From Foundations to ML, DL, NLP, GenAI, Agentic AI & MLOps.

Trainer
Srinivas Dande



**Dandes
Academy**

List of Modules

Course 0: SQL, Advanced SQL and Data Modeling(Self Learning)

Course 1: Python Fundamentals

Course 2: Python for ML & DS

Course 3: Mathematics for ML & DS

Course 4: Exploratory Data Analysis (EDA)

Course 5: Machine Learning - Foundation

Course 6: Advanced Machine Learning

Course 7: Deep Learning - Foundation

Course 8: Advanced Deep Learning

Course 9: Image Processing & Computer Vision

Course 10: Natural Language Processing (NLP)

Course 11: Generative AI

Course 12: Large Language Models (LLMs) & Fine-Tuning

Course 13: Agentic AI (NEW)

Course 14: MLOps

Course 15: End-to-End AI/ML Project



Course 0: SQL, Advanced SQL, and Data Modeling(Self)

- **Topic Coverage**

- ✓ SQL Query Fundamentals
- ✓ Set Operators
- ✓ Group By & Aggregations
- ✓ Joins
- ✓ Common Table Expressions (CTEs)
- ✓ Window Functions
- ✓ RANK, ROW_NUMBER, LEAD, LAG, etc
- ✓ Nested Queries
- ✓ Indexes
- ✓ Query Optimization & Best Practices
- ✓ Data Modeling (OLTP vs OLAP)
- ✓ Entity Relationships
- ✓ Normalization
- ✓ Star Schema & Snowflake Schema

- **Assessment**

- ✓ Practice Questions
- ✓ Assignment #1 - Advanced SQL
- ✓ Assignment #2 - Data Modeling
- ✓ Test 3 - SQL, Advanced SQL, and Data Modeling



Course 1: Python Fundamentals

- **Topic Coverage**

- ✓ Data Types & Variables
- ✓ Conditional Statements (if-else)
- ✓ Loops (for, while)
- ✓ Functions & Lambda Functions
- ✓ Strings, Lists, Tuples, Sets, Dictionaries
- ✓ Comprehensions (List, Dict, Set)
- ✓ Functional Programming
- ✓ Object-Oriented Programming
- ✓ Decorators
- ✓ Generators & Iterators
- ✓ Exception Handling
- ✓ File Handling

- **Assessment**

- ✓ Practice Questions
- ✓ Assignment (Python Problem Solving)
- ✓ Test 1 - Python Fundamentals



Course 2: Python for ML & DS

- **Topic Coverage**

- ✓ NumPy
- ✓ Pandas
- ✓ Matplotlib
- ✓ Seaborn

- **Assessment**

- ✓ Practice Questions
- ✓ Assignment #1 - Pandas
- ✓ Assignment #2 - Plotting
- ✓ Test 2 - Python for ML & DS



Course 3: Mathematics for ML & DS

- **Topic Coverage**

- ✓ Probability Basics & Rules
- ✓ Probability Distributions
- ✓ Important Theorems
- ✓ Descriptive & Inferential Statistics
- ✓ Hypothesis Testing
- ✓ Correlation & Covariance
- ✓ Linear Algebra
- ✓ Vectors, Matrices
- ✓ Eigenvalues & Eigenvectors
- ✓ Matrix Decomposition
- ✓ Calculus for ML
- ✓ Derivatives & Gradients
- ✓ Optimization Intuition
- ✓ Regression Mathematics (Linear & Polynomial)

- **Assessment**

- ✓ Practice Questions
- ✓ Assignment (Math → ML Mapping)
- ✓ Test 4 - Mathematics for ML & DS



Course 4: Exploratory Data Analysis (EDA)

- **Topic Coverage**

- ✓ Understanding the Data
- ✓ Data Cleaning & Validation
- ✓ Handling Missing Data & Outliers
- ✓ Feature Engineering
- ✓ * Encoding Techniques
- ✓ * Scaling & Normalization
- ✓ * Feature Transformation
- ✓ * Time-based Features
- ✓ Dimensionality Reduction (PCA, VIF)
- ✓ Data Leakage Detection
- ✓ Model Evaluation Metrics
- ✓ Train / Validation / Test Split
- ✓ Stratified Sampling
- ✓ Data-Centric AI Principles

- **Assessment**

- ✓ Practice Questions
- ✓ Assignment (EDA + Feature Engineering Project)
- ✓ Test 5 - Exploratory Data Analysis (EDA)



Course 5: Machine Learning - Foundation

- **Topic Coverage**

- ✓ Introduction to Machine Learning
- ✓ Supervised vs Unsupervised Learning
- ✓ Supervised Learning
- ✓ Regression Algorithms
 - Linear Regression
 - Regularization (L1, L2)
- ✓ Classification Algorithms
 - Logistic Regression
 - Naïve Bayes
 - K-Nearest Neighbors (KNN)
 - Support Vector Machines (SVM)
 - Handling Imbalanced Data
- ✓ Unsupervised Learning
- ✓ Clustering Techniques
 - K-Means
 - Hierarchical Clustering
 - Elbow Method
 - Clustering Metrics & Evaluation
- ✓ Evaluation Metrics
 - Regression Metrics
 - Classification Metrics

- **Assessment**

- ✓ Practice Questions
- ✓ Assignment - Supervised Learning
- ✓ Assignment - Unsupervised Learning
- ✓ Test 6 - Machine Learning - Foundation



Course 6: Advanced Machine Learning

- **Topic Coverage**

- ✓ Advanced Supervised Learning
- ✓ Decision Trees
- ✓ Bagging Techniques
- ✓ Random Forest
- ✓ Boosting Techniques
 - AdaBoost
 - Gradient Boosting
 - XGBoost
- ✓ Advanced Unsupervised Learning
- ✓ Advanced Clustering
 - DBSCAN
 - Birch
 - GMM
- ✓ Expectation Maximization
- ✓ Anomaly Detection
- ✓ Dimensionality Reduction
- ✓ Model Selection & Optimization
- ✓ Time Series Forecasting

- **Assessment**

- ✓ Practice Questions
- ✓ Mini Project #1 - Supervised Learning
- ✓ Mini Project #2 - Unsupervised Learning
- ✓ Test 7 - Advanced Machine Learning



Course 7: Deep Learning - Foundation

- **Topic Coverage**

- ✓ Deep Learning Fundamentals
- ✓ Basics of Neural Networks
- ✓ Forward & Backpropagation
- ✓ Gradient Descent & Optimization
- ✓ Activation Functions
 - Sigmoid, tanH, ReLU
 - Leaky ReLU, Softmax
- ✓ Training Deep Neural Networks
- ✓ Weight Initialization Strategies
- ✓ Regularization Techniques
 - L1, L2, Dropout
 - Batch Normalization
- ✓ Learning Rate Schedulers
- ✓ Optimizers - SGD, Adam, RMSprop-
- ✓ Tensors & Computation
- ✓ Deep Learning Frameworks
 - PyTorch and TensorFlow
- ✓ Core Deep Learning Architectures
 - Convolutional Neural Networks (CNNs)
 - Recurrent Neural Networks (RNNs)
 - Long Short-Term Memory (LSTMs)
- ✓ Hyperparameter Tuning & Model Optimization

- **Assessment**

- ✓ Practice Questions
- ✓ Assignment
- ✓ Test 8 - Deep Learning - Foundation



Course 8: Advanced Deep Learning

- **Topic Coverage**

- ✓ Advanced Deep Learning Architectures
 - Advanced CNN Architectures
 - RNN, LSTM, GRU
 - Transfer Learning
- ✓ Advanced Training & Optimization
 - Deep Learning Model Optimization
 - Training Stability Issues
 - Cross-Validation in Deep Learning
 - Hyperparameter Search
- ✓ Model Explainability & Responsible AI
 - Model Interpretability - SHAP and LIME
 - AI Fairness & Bias Reduction
- ✓ Transformers
 - Introduction to Transformers
 - High-Level Overview of Transformer Architectures

- **Assessment**

- ✓ Practice Questions
- ✓ Mini Project #3 : Deep Learning Applications
- ✓ Test 9: Advanced Deep Learning



Course 9: Computer Vision

- **Topic Coverage**

- ✓ Image Processing Fundamentals
- ✓ Convolutional Neural Networks for Vision
- ✓ Fundamentals of CNNs
- ✓ Modern CNN Architectures
 - SqueezeNet
 - ResNet
 - EfficientNet
- ✓ Image Classification
 - Image Classification Concepts
 - Pre-trained CNN Models
 - Custom CNN Models
- ✓ Object Detection
 - Object Detection Fundamentals
 - YOLO, SSD, Faster R-CNN
- ✓ Image Augmentation & Preprocessing
 - Image Augmentation Techniques
 - Data Preprocessing for Vision

- **Assessment**

- ✓ Practice Questions
- ✓ Mini Project #4 : Image Classification
- ✓ Mini Project #5: Real-Time Object Detection
- ✓ Test 10: Computer Vision



Course 10: Natural Language Processing (NLP)

- **Topic Coverage**

- ✓ NLP Fundamentals
 - Introduction to NLP
 - Text Preprocessing & Tokenization
- ✓ Text Representation
 - Word Embeddings
 - Word2Vec, GloVe, FastText
- ✓ Deep Learning for NLP
 - * Neural Networks for NLP
 - * Seq2Seq Models
 - * Transformers for NLP
- ✓ Pretrained Language Models
 - BERT
 - T5
 - Other Pretrained Models
- ✓ NLP Applications
 - Named Entity Recognition (NER)
 - Sentiment Analysis
 - Sentiment Analysis on Product Reviews

- **Assessment**

- ✓ Practice Questions
- ✓ Mini Project #6 : AI Chatbot for Customer Support
- ✓ Test 11: Natural Language Processing (NLP)



Course 11: Generative AI

- **Topic Coverage**

- ✓ Introduction to Generative AI
- ✓ Applications in Text, Image, and Video Generation
- ✓ Autoencoders
 - Introduction to Autoencoders
 - Variational Autoencoders (VAEs)
 - Denoising Autoencoders
 - Sparse Autoencoders
- ✓ Generative Adversarial Networks (GANs)
 - GAN Fundamentals
 - GAN Variants
- ✓ Diffusion Models
 - Diffusion Models Overview
 - Stable Diffusion
 - DALL·E
 - MidJourney
- ✓ Generative AI Applications
 - AI-Generated Art using GANs
 - Image-to-Image Translation with CycleGAN
 - Text-to-Image Generation with Stable Diffusion

- **Assessment**

- ✓ Practice Questions
- ✓ Mini Project #7 : AI-Generated Art & Image-to-Image Translation
- ✓ Test 12: Generative AI



Course 12: LLMs & Fine-Tuning

- **Topic Coverage**

- ✓ Introduction to Large Language Models
- ✓ Evolution of LLMs- GPT, BERT, T5, LLaMA, Claude, Gemini
- ✓ LLM Foundations
 - Tokenization Strategies
 - WordPiece, SentencePiece, Byte Pair Encoding (BPE)
 - Pretraining vs Fine-Tuning vs Prompt Engineering
- ✓ LLM Frameworks & Tooling
 - LangChain and LlamaIndex
- ✓ Fine-Tuning Large Language Models
 - Fine-Tuning Open-Source LLMs - LLaMA, Falcon
 - Parameter-Efficient Fine-Tuning (PEFT) and LoRA
- ✓ Alignment & Optimization
 - Reinforcement Learning with Human Feedback (RLHF)
 - Reward Models and Multi-Armed Bandits
 - Proximal Policy Optimization (PPO) for LLMs
- ✓ Retrieval-Augmented Generation (RAG)
 - RAG Architecture & Workflow
 - Building RAG Pipelines for LLMs
- ✓ LLM Applications
 - Building AI Chatbots & Assistants
 - Code Generation using LLMs- CodeLlama, StarCoder
 - Text Summarization and AI-Powered Content Creation

- **Assessment**

- ✓ Practice Questions
- ✓ Mini Project #8 : Fine-Tuning an LLM for a Domain-Specific Chatbot
- ✓ Test 13: LLMs & Fine-Tuning



Course 13: Agentic AI (Autonomous & Multi-Agent Systems)

- **Topic Coverage**

- ✓ Introduction to Agentic AI
 - What is Agentic AI
 - LLMs as Reasoning Agents
 - Agentic AI vs Traditional AI Pipelines
 - Single-Agent vs Multi-Agent Systems
- ✓ Core Agent Concepts
 - Agent Architecture
 - Perception, Planning, Reasoning, Action Loop
 - Tool Calling & Function Calling
 - Prompting Strategies for Agents
- ✓ Memory in Agents
 - Short-Term Memory
 - Long-Term Memory
 - Conversation Memory
 - Vector Memory using Embeddings
- ✓ Planning & Reasoning
 - Task Decomposition
 - Planning Strategies (ReAct, Plan-and-Execute)
 - Chain-of-Thought & Reasoning Patterns
 - Handling Failures & Retries
- ✓ Agent Frameworks & Tooling
 - LangChain Agents
 - LlamaIndex Agents
 - Tool Integration (APIs, Databases, Web Tools)
- ✓ Multi-Agent Systems
 - Multi-Agent Architectures
 - Agent Collaboration & Coordination
 - Role-Based Agents
 - Task Delegation among Agents

- ✓ Autonomous AI Agents
 - Autonomous Decision Making
 - Long-Running Agents
 - Event-Driven Agents
 - Human-in-the-Loop Systems
- ✓ Agent Evaluation, Safety & Control
 - Agent Evaluation Metrics
 - Guardrails & Safety Mechanisms
 - Preventing Hallucinations
 - Cost & Latency Optimization
 - Ethical Considerations in Agentic AI
- ✓ Agentic AI Applications
 - AI Personal Assistants
 - Autonomous Research Agents
 - Customer Support AI Agents
 - Data Analysis & Reporting Agents
- **Assessment**
 - ✓ Practice Questions
 - ✓ Mini Project #9: Build an Autonomous AI Agent
 - ✓ Test 14: Agentic AI



Course 14: MLOps

- **Topic Coverage**
 - ✓ ML Lifecycle & Foundations
 - End-to-End ML Model Lifecycle
 - MLOps Best Practices
 - Experiment Tracking & Reproducibility
 - ✓ Model & Feature Management
 - Model Versioning
 - Feature Store Management
 - Data Versioning
 - ✓ MLOps Tooling & Platforms
 - MLflow, Kubeflow
 - TensorFlow Extended (TFX)
 - Kubernetes for Machine Learning
 - ✓ CI/CD for Machine Learning
 - CI/CD Pipelines for ML
 - GitOps for ML
 - Kubeflow Pipelines
 - Automated Training & Deployment
 - ✓ Model Deployment
 - Model Serving & Inference
 - Online vs Batch Inference
 - ✓ Monitoring, Evaluation & Optimization
 - Model Monitoring
 - Data Drift vs Concept Drift
 - Model Performance Tracking
- **Assessment**
 - ✓ Practice Questions
 - ✓ Assignment
 - ✓ Test 15: MLOps



Course 15: Final End-to-End AI/ML Project

- **Topic Coverage**
 - ✓ **Capstone Project**
 - AI-Powered Personal Assistant (Multi-Modal AI Agent)
 - ✓ **Project Scoping & Design**
 - Problem Definition
 - Requirements Gathering
 - System Architecture Design
 - ✓ **Data Collection & Preprocessing**
 - Data Collection, Cleaning & Transformation
 - Dataset Versioning
 - ✓ **Feature Engineering & Selection**
 - Feature Engineering
 - Feature Selection Techniques
 - ✓ **Model Training & Optimization**
 - Model Selection
 - Training Pipelines
 - Hyperparameter Optimization
 - ✓ **Model Evaluation & Validation**
 - Evaluation Metrics
 - Validation Strategies and Error Analysis
 - ✓ **Model Deployment**
 - API Deployment
 - Web Application Deployment
 - ✓ **Model Monitoring & Maintenance**
 - Model Monitoring
 - Performance Tracking and Drift Detection
 - ✓ **Scaling & Continuous Improvement**
 - Scalability Strategies and Cost Optimization
 - Continuous Model Improvement



Course Overview:

- ✓ This course provides a comprehensive, industry-aligned learning path for mastering Artificial Intelligence and Machine Learning. Designed with real-world applications and modern best practices, the program equips learners with the skills required to build, deploy, and optimize AI/ML models.
- ✓ Whether you are aiming for a career in AI, Data Science, or Machine Learning Engineering, this course offers a structured, hands-on approach to mastering AI concepts — from core fundamentals to advanced techniques.
- ✓ **This course is taught step-by-step. No prior AI/ML experience is required.**

Who Should Attend:

- ✓ **Software Engineers & Developers**
 - Professionals looking to transition into AI/ML, Data Science, or AI Engineering roles.
- ✓ **Data Scientists & ML Engineers**
 - Practitioners who want to upskill with advanced AI/ML, Generative AI, LLMs, Agentic AI, and MLOps concepts.
- ✓ **Tech Professionals & AI Enthusiasts**
 - Individuals aiming to break into the AI/ML industry with a structured, hands-on learning approach.
- ✓ **Professionals Preparing for AI/ML Interviews**
 - Candidates preparing for AI/ML roles at top product-based companies through strong fundamentals, projects, and interview-focused preparation.
- ✓ **Job Seekers Targeting High-Growth AI/ML Roles**
 - Learners aiming for high-impact and high-growth AI/ML roles in leading technology-driven organizations.

Why Dandes Academy?

- ✓ Choosing the right AI/ML program is not just about learning algorithms — it's about building real, industry-ready skills that translate into long-term career growth.
- ✓ At Dandes Academy, we focus on depth, structure, and real-world application, not shortcuts.

1) Industry-Aligned, Future-Ready Curriculum

- Our 16-module AI/ML program is carefully designed to take you from foundations to advanced, production-level AI systems.
- You don't just learn models — you learn how to:
 - ✓ Build end-to-end AI/ML solutions
 - ✓ Work with Generative AI, LLMs, Agentic AI
 - ✓ Deploy and manage models using MLOps best practices
- This ensures you're ready not just for today's roles, but for future AI careers.

2) Step-by-Step Learning (Beginner Friendly)

- This course is taught step-by-step.
 - ✓ No prior AI/ML experience is required.
- **We start with:**
 - ✓ Python, SQL, Data Modeling
 - ✓ Mathematics, Statistics, EDA
 - ✓ And gradually move into:
 - ✓ Machine Learning → Deep Learning
 - ✓ NLP → Generative AI → LLMs
 - ✓ Agentic AI → MLOps → Capstone Project
- This structured approach ensures clarity, confidence, and strong fundamentals.

3) Founder-Led Teaching with 20+ Years of Experience

- This program is taught by **Srinivas Dande, Founder & Instructor at Dandes Academy**, with **20+ years of industry and teaching experience**.
- **What this means for you:**
 - ✓ Clear explanations from basics to advanced
 - ✓ Real-world insights, not just theory
 - ✓ Guidance aligned with actual industry expectations

4) Hands-On Learning with Real Projects

- Learning happens by doing, not just watching.
- **You will work on:**
 - ✓ Multiple mini-projects
 - ✓ Real-time industry-oriented projects
 - ✓ A final end-to-end AI/ML capstone project
- This builds a strong project portfolio that recruiters value.

5) Advanced Skills That Most Courses Don't Teach

- **Unlike many programs, we go deep into:**
 - ✓ Generative AI & LLMs
 - ✓ Fine-tuning & RAG pipelines
 - ✓ Agentic AI (Autonomous & Multi-Agent Systems)
 - ✓ Production MLOps & deployment
- These are the skills companies are actively looking for right now.

6) Strong Career & Placement Focus

- We don't just teach — we prepare you for the job market.
- **You get:**
 - ✓ Placement assistance
 - ✓ Resume & interview guidance
 - ✓ Mock interviews
 - ✓ Internship opportunities (based on performance)
- This helps you confidently target roles such as:
 - ✓ ML Engineer,
 - ✓ AI Engineer,
 - ✓ Gen AI Engineer,
 - ✓ Data Scientist,
 - ✓ Lead ML Engineer,
 - ✓ ML Architect.

7) Long-Term Learning & Flexibility

- 12 months structured program
- 5 years LMS access for recordings & materials
- Batch switching option (subject to availability)
- 100% refund within 30 days if you're not satisfied — no questions asked



What You Will Learn:

- ✓ **Strong AI & ML Foundations**
 - Master core concepts in Supervised and Unsupervised Learning, along with Statistics, Mathematics for ML, and Exploratory Data Analysis (EDA) to build a solid foundation.
- ✓ **End-to-End Machine Learning**
 - Design, train, evaluate, and optimize Machine Learning models using industry-standard techniques, including feature engineering, model selection, and hyperparameter tuning.
- ✓ **Deep Learning & Neural Networks**
 - Gain hands-on experience with Deep Learning architectures, including ANNs, CNNs, and RNNs, and apply them to real-world problems.
- ✓ **Natural Language Processing (NLP)**
 - Work with text data, covering text preprocessing, embeddings, transformers, and NLP use cases.
- ✓ **Generative AI & Large Language Models (LLMs)**
 - Explore Generative AI, including Transformers, Diffusion Models, LLMs, and fine-tuning techniques, and build practical GenAI applications.
- ✓ **Agentic AI Systems**
 - Learn to design and build Agentic AI workflows, including multi-step reasoning, tool usage, and real-world agent-based applications.
- ✓ **MLOps & Model Deployment**
 - Build end-to-end AI pipelines using MLOps best practices, including model deployment, versioning, monitoring, and automation.
- ✓ **Hands-On Projects & Real-Time Case Studies**
 - Work on multiple mini-projects and real-time industry projects to gain practical exposure and build a strong project portfolio.
- ✓ **Industry-Ready AI Skills**
 - Gain practical skills to design, deploy, and scale AI/ML solutions, preparing you for real-world roles such as ML Engineer, AI Engineer, and ML Architect.